Siqiao Mu

U.S. Citizen
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https://siqiaomu.github.io/

Education

Northwestern University — Evanston, IL
Ph.D. in Engineering Sciences and Applied Mathematics — 2021–2027
Advisor: Diego Klabjan, Industrial Engineering and Management Sciences

California Institute of Technology — Pasadena, CA
B.S. in Applied and Computational Mathematics — 2016–2020

Research Interests

Deep learning theory, nonconvex optimization, stochastic optimization, gradient algorithms, AI safety, differential privacy, machine unlearning, low-rank training, reinforcement learning

Publications

[2] S. Mu and D. Klabjan, "On the convergence rate of LoRA gradient descent", *Forty-third International Conference on Machine Learning*, 2026, [link]. **Oral presentation (top 0.7%)**.

[4] S. Mu and D. Klabjan, "Rewind-to-delete: Certified machine unlearning for nonconvex functions", *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025, [link].

[5] S. Mu and D. Klabjan, "On the second-order convergence of biased policy gradient algorithms", *Forty-first International Conference on Machine Learning*, 2024, [link].

[6] M. Ahlers, M. Bustamante, and S. Mu, "Unitarity bounds of astrophysical neutrinos", *Physical Review D*, vol. 98, no. 12, 2018, [link].

Preprints

[1] S. Mu and D. Klabjan, *Convergence guarantees of gradient descent for neural networks via generalized Lipschitz smoothness*, 2026. arXiv: 2608.11479 [cs.LG].

[3] S. Mu and D. Klabjan, *Descend or rewind? Stochastic gradient descent unlearning*, 2025. arXiv: 2511.15983 [cs.LG].

Teaching

- MLDS 490/IEMS 469: Advanced Algorithms for Machine Learning — Fall 2024, Fall 2025, Fall 2026
  Developing course materials, holding office hours, and grading, as the main TA for a class of ∼25 graduate students
  Project topics: Policy Gradient, Deep Q-Networks, Federated Learning, Automated Feature Engineering
- MLDS 420: Machine Learning II — Winter 2026
  Developing slides, leading lab sessions and exam review sessions, grading
- MLDS 421: Data Mining & Knowledge Discovery — Winter 2026
  Developing slides, leading lab sessions, grading
- MLDS 410: Generating Business Value with Analytics — Winter 2024
  Grading for a class of ∼20 graduate students

- MATH 228: Multi-variable Calculus — Spring 2023
  Leading lab sessions, office hours, grading, as one of 2 TAs for a class of ∼80 undergraduate students
- EA 4: Differential Equations — Fall 2022
  Leading lab sessions, office hours, grading, as one of 4 TAs for a class of ∼100 undergraduate students
- ACM 116: Introduction to Probability Models (Caltech) — Fall 2019
  Office hours, grading, as one of 5 TAs for a class of ∼100 undergraduate students

Leadership and Service

- Institute of Industrial and Systems Engineers (IISE) Annual Conference & Expo — 2026
  Modeling & Simulation Track Session Chair
- Forty-third International Conference on Machine Learning — 2026
  Silver Reviewer (Top 50%)
- IEEE Transactions on Neural Networks and Learning Systems — 2025
  Reviewer
- The Thirty-Ninth Annual Conference on Neural Information Processing Systems — 2025
  Reviewer
- Society for Industrial and Applied Mathematics (SIAM) Northwestern Student Chapter — 2023–2025
  President
  Responsibilities: Organize meetings and events, coordinate payments, recruit members, recruit volunteers to help with events, interface with administration and other SIAM Chapters
  Some events:Chicago Area SIAM Student Conference (CASSC25), Argonne National Laboratory Field Trip, collaborative events with INFORMS, weekly SIAM Journal Club, Undergrad Boba Networking
- Northwestern M.S. in Machine Learning and Data Science Program — Winter 2025
  Admissions application reviewer
- Northwestern Women in Mathematics (WIM) — 2023–2025
  Member

Other Conferences

- INFORMS Annual Meeting (San Francisco, CA) — 2026
  Job Market Showcase: "Convergence analysis of low-rank training in machine learning"
- IISE Annual Conference & Expo (Arlington, Texas) — 2026
  Invited talk: "Stochastic gradient descent algorithms for machine unlearning"
- INFORMS Annual Meeting (Atlanta, Georgia) — 2025
  Invited talk: "Gradient-based methods for machine unlearning"
- SIAM Annual Meeting (Montreal, Canada) — 2025

Awards

- NSF IISE Student Travel Award — 2026
  Support for students to participate in the 2026 Institute of Industrial and Systems Engineers (IISE) Annual Conference and Expo.
- Rising Stars in Computational & Data Sciences (hosted at the Santa Fe Institute) — 2026
  "Nominees are considered based on their technical merit, and their technical and service leadership."
- Northwestern Presidential Fellowship Nominee — 2025
  Sole nominee from the ESAM department

- SIAM Student Chapter Certificate of Recognition 2024
  "Acknowledging Siqiao Mu for exceptional service to the Northwestern University Student Chapter during the 2023-2024 academic year"
- NSF Graduate Research Fellowship Program Honorable Mention 2023

Skills

Python (PyTorch, TensorFlow, Numpy, Pandas), MATLAB, C, C++, SQL, high-performance computing, bash, Kubernetes, JavaScript, R

Additional Experience

Los Alamos National Laboratory Los Alamos, NM
Cybersecurity Summer School Research Intern Summer 2024
Mentor: Juston Moore
Out-of-of distribution detection via neural tangent kernel methods

Nordstrom Seattle, WA
Data Science Intern Summer 2021
Combining prompt engineering with GPT-3 and UI/UX design to create a tool for generating clothing product descriptions

UCLA/Caltech Pasadena, CA
Summer Undergraduate Research Fellowship Summer 2020
Mentors: Franca Hoffmann, Mason Porter, Heather Zinn-Brooks
Utilizing mean-field approximations to model the effect of mixing patterns on disease spread

BlackRock New York, NY
Summer Analyst (Portfolio Analytics Group) Summer 2019

Niels Bohr Institute Copenhagen, DNK
Summer Undergraduate Research Fellowship Summer 2018
Mentors: Markus Ahlers and Mauricio Bustamante
Modeling astrophysical neutrino flavor oscillations based on new physics theories

NYU MRSEC New York, NY
Research Experience for Undergraduates Summer 2017
Mentor: David Grier